GAD2003-N – Game Engine Construction Report

Cassius Lockwood – E4101143

User Guide

The game is a Real Time Strategy, with an objective of collecting as many resources as possible. The view is top-down, and the game supports 2 structure types, being a power plant and a drill, for resource collection. The only way to lose the game is by not having enough resources to continue.

Controls

W	Camera Up
A	Camera Left
S	Camera Down
D	Camera Right
Left Mouse Button	Place a Structure
E	Switch Structure to Place.

Structure guide

Structure	Cost - Resources	Cost - Power	Production - Resources	Production - Power
Power Plant	100	0	0	10
Drill	200	10	100/10s	0

Known issues

The main and only known bugs with regards to gameplay, is a lack of an official end of game screen, and no text for resources or power.

This would have been fixed however time constraints became limiting. The above issues can be solved by using the provided structure table, which will apply to each structure. This will require the player to work out resource production, power consumption, etc.

There is one technical bug, which either results in a memory leak or an error saying the memory being deleted wasn't valid. I was unable to fix this mostly due to time constraints, but also due to not being able to find the source of the issue.

Implementation checklist

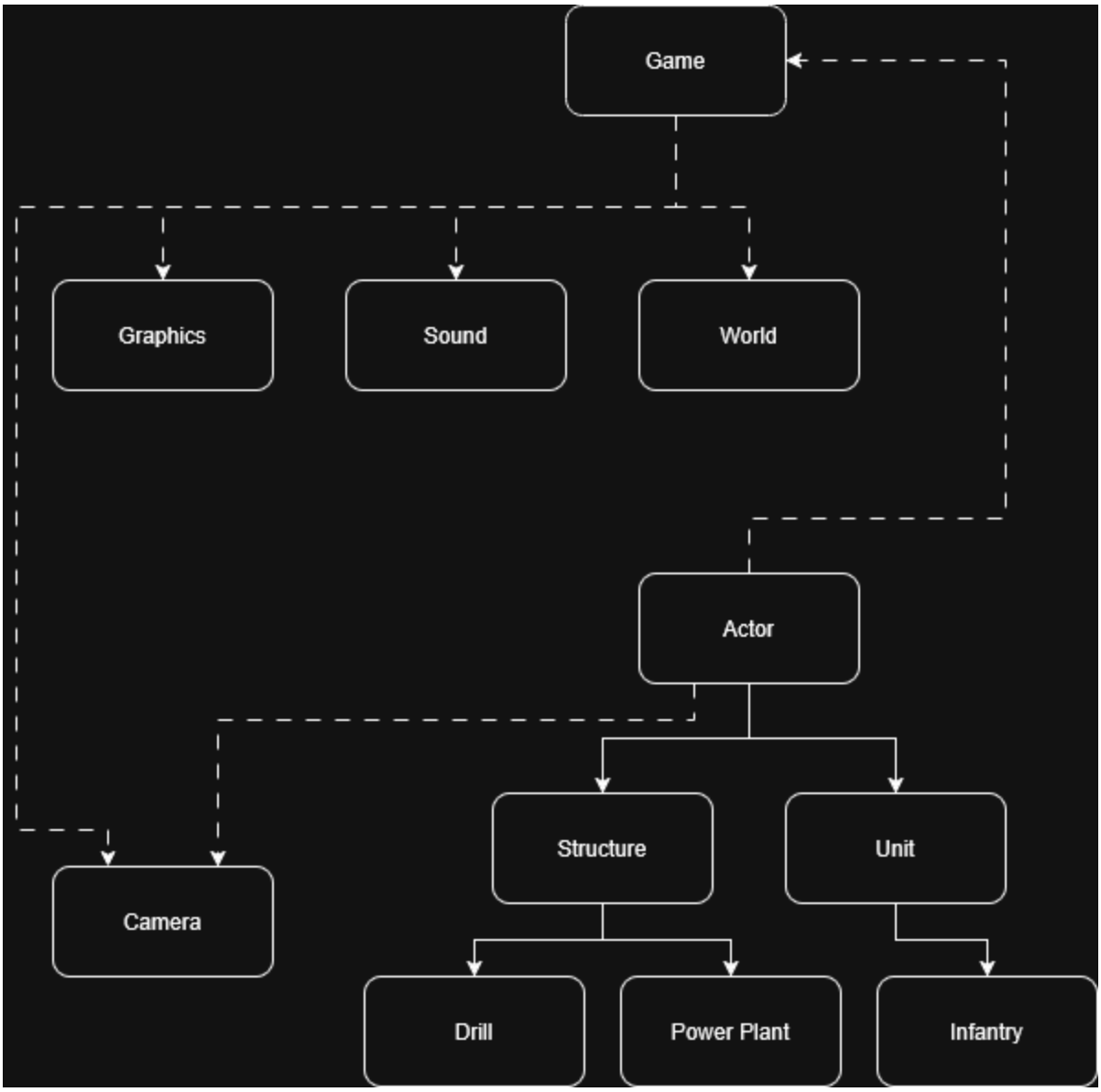
I was sadly unable to directly import the full checklist, so all of the complete features are:

- Additional engine support, i.e. AI or sound

- Mapping world space onto screen space
- More advanced use of C++ i.e. namespaces, STL, C++ 11, libraries and patterns.
- Patterns correctly chosen and implemented.
- Systems are easily maintainable and be able to be extended
- Good error handling throughout
- Basic AI routines, such as paths, state machines, etc.
- Gameplay elements are evident
- Class function and member variable visibility is correct
- There are multiple world entity types
- Report completed.
- Memory is only managed outside of game loop (allocation / de-allocation)
- Code is well commented
- Code can be built and executed without errors or warnings.
- Interpolation to smooth enemy movement (delta time)
- Player entity representation fully working
- Debug is supported via debug output and / or asserts.
- Fully working Graphics System is in place
- Multiple animation sets are supported
- Working simulation of the game world in place

Maintenance Guide

The XML guide is as follows:



Dashed lines represent where something is used, for example, Game.h having the WorldGraphics, which is an instance of Graphics. WorldGraphics is then passed in as a reference to others, like Actor, but Actor does not make a new instance of Graphics, so is not included as a trace in the XML diagram.

To note, AnimDataStruct and GameData are not included in this diagram, as they are simply used as references, but are included in a majority of the design.

More features were planned, and some code was in place, but were never added due to time constraints, such as the Infantry, sound, and Colliders.

Conclusion,

Sadly, I had to pivot my focus towards my other module towards the end of the semester, due to a number of software issues. This led me to not get as much as I had hopped in the engine, but that I had structure for, like AI, projectiles, colliders working, physics, etc. I feel like the scope of the game was also too large, considering the other module and other commitments I had. I feel if I was able to put more time in, I would have achieved the product I wanted, but simply ran out of time.

I have also learnt a great deal regarding the design of code, rather than writing the code, and that the design of the code begins to outweigh the short-term efficiency of the code, so long-term planning should always be completed, or at least considered, before adding any new features or elements to the project.

If I was to continue making this game, I would focus on finalizing the features I tried to add but ran out of time for, and adding a proper game loop, such as win conditions, and display of game data like resources, power, etc. With that being said, I am quite happy with the state of my project, and especially happy with the knowledge I have gained from this project.